Amine NAKROU

Student Engineer – Computer Science, Statistics & AI

Polytech Lille — ISIA 4th year

4-month Internship • From April 14, 2026

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Profile

4th-year student engineer (ISIA) at Polytech Lille, passionate about **Data Science** and **Machine Learning** — fields in which I actively invest through my academic and personal projects. Curious, rigorous, and always seeking new challenges, I am looking for a **4-month internship** starting **April 14, 2026** to apply my skills to concrete and ambitious projects.

Education

2024 – 2027 Polytech Lille

Engineering Cycle ISIA

ML, Data Mining, SQL/NoSQL, Markov Models, Statistics, Algorithms, Operations Research, IS Architecture & Big Data, Project Management.

2022 – 2024 Lycée Moulay Idriss, Fès

Preparatory Classes MPSI/MP

Algebra, Analysis, Algorithms, Python, Statistics.

2022 Taounate, Morocco

Baccalaureate in Mathematical Sciences

Graduated with **Highest Honors**.

Skills

Languages: Python, R, Java, C/C++, SQL, JavaScript, HTML, CSS, PHP

ML / Data: scikit-learn, XGBoost, PyTorch, Pandas, NumPy, Streamlit, MLflow

Big Data / IS: Hadoop, Spark (basics), IS Architecture, Data Pipelines

Databases: MySQL, PostgreSQL, MongoDB, NoSQL

Tools: Docker, Git, Power BI, Tableau, LaTeX, MS Office

Methods: Agile/Scrum, Project Management, Data Valorization

Languages: French (fluent) · English (C1) · Arabic (native) · Spanish (A2)

Professional Experience

Data & Web Developer Intern — HAMATEL S.A.R.L *Summer 2025 | 2 months | Morocco*

- Designed a **relational database** (modeling, normalization, SQL) and developed an **internal web application** (Flask) for data management and visualization.
- Performed exploratory analysis and **data preparation** to produce decision-support reports.

Academic & Personal Projects

Customer Churn Prediction — End-to-End ML Pipeline *Python · XGBoost · Streamlit · Docker · MLflow*

- **Feature engineering** and model comparison (Random Forest, XGBoost) on imbalanced tabular data; achieved **AUC 0.91** after hyperparameter tuning.
- Deployed a **containerized Flask API** (Docker) serving the model in production, interactive **Streamlit** dashboard for prediction visualization, and experiment tracking via **MLflow**.

EA Sports FC (FIFA) Project — Data Analysis & Exploitation *Python · Pandas · Matplotlib · Scikit-learn · Jupyter*

- Cleaned and explored a FIFA dataset (2016–2024) with **Pandas**: salary/performance correlation analysis, outlier detection, and statistical profiling of the top 10 players.
- Trained **explainable classification models** (decision trees, logistic regression) to predict performance categories; interpreted results through feature importance analysis.

Time Project — Software Development Effort Estimation *R · RMarkdown · Linear Regression · Supervised Classification · Advanced Modeling*

- Modeled development effort on AT&T projects (function points, OS, DBMS) using **multiple linear regression**: variable selection, assumption verification (normality, homoscedasticity), and residual analysis.
- Applied **supervised classification methods** (k-NN, decision tree) and **cross-validation** to assess model robustness; produced an interactive **RMarkdown** report with dynamic visualizations.

Production Optimization *Linear Programming · Lindo · Mathematical Modeling*

- Modeled a **linear programming (LP) problem** to maximize profit for a perfume manufacturing company: defined decision variables and formalized resource and production constraints.
- Solved exactly using **Lindo** and performed sensitivity analysis: interpreted slack variables, dual prices, and stability ranges to guide operational decisions.

Interests

Football · Swimming · Travel & cultural discovery · New technologies & AI